BS-M301 — Probability and Statistics

Semester III • 3L:0T:3P • 3 Credits

Module 1 Exam Notes: Basic Probability, Discrete Random Variables and Inequalities

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Probability Spaces, Conditional Probability and Independence

Modern probability theory is founded upon the rigorous measure-theoretic framework introduced by Andrey Kolmogorov in 1933. Rather than relying on heuristic ratios or subjective belief, probability is defined as a normalized measure assigned to events defined on a sample space.

Definition: Probability Space $(\Omega, \mathcal{F}, P)$

A **probability space** is a formal mathematical triplet $(\Omega, \mathcal{F}, P)$ comprising:

1. **Sample Space (Ω):** The set of all possible fundamental outcomes ω of a random experiment ($\omega \in \Omega$).
2. **Event Space (σ-algebra $\mathcal{F}$):** A collection of subsets of Ω satisfying three closure properties:
   - $\Omega \in \mathcal{F}$ (the certain event is included).
   - If $A \in \mathcal{F}$, then its complement $A^c = \Omega \setminus A \in \mathcal{F}$ (closure under complementation).
   - If $A_1, A_2, A_3, \cdots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (closure under countable unions).
3. **Probability Measure (P):** A real-valued mapping $P : \mathcal{F} \to [0, 1]$ satisfying Kolmogorov's Axioms.

Kolmogorov's Axioms of Probability

Every valid probability measure P on $(\Omega, \mathcal{F})$ must strictly obey:

$$\textbf{Axiom 1 (Non-negativity):} \quad P(A) \geq 0, \quad \forall A \in \mathcal{F} \tag{1}$$

$$\textbf{Axiom 2 (Total Certainty):} \quad P(\Omega) = 1 \tag{2}$$

$$\textbf{Axiom 3 (Countable Additivity):} \quad P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i), \quad \text{if } A_i \cap A_j = \emptyset \quad (\forall i \neq j) \tag{3}$$

1.1 Fundamental Theorems Derived from Kolmogorov's Axioms

From these three primitive axioms, standard probability laws follow deductively:

1. **Empty Set Probability:** $P(\emptyset) = 0$, since $\Omega \cup \emptyset = \Omega$ and $\Omega \cap \emptyset = \emptyset \implies P(\Omega) + P(\emptyset) = P(\Omega) \implies P(\emptyset) = 0$.
2. **Complement Law:** For any event A, $A \cup A^c = \Omega$ and $A \cap A^c = \emptyset$, hence:

$$P(A^c) = 1 - P(A) \tag{4}$$

3. **Monotonicity:** If $A \subseteq B$, then $B = A \cup (B \setminus A)$ with $A \cap (B \setminus A) = \emptyset$. Thus $P(B) = P(A) + P(B \setminus A) \geq P(A)$, establishing:

$$A \subseteq B \implies P(A) \leq P(B) \tag{5}$$

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4. **Addition Theorem for Arbitrary Events (Inclusion-Exclusion):** For any two events A and B:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{6}$$

For three events A, B, C:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C) \tag{7}$$

5. **Boole's Inequality (Union Bound):** For any countable collection of events $A_1, A_2, \ldots$:

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i) \tag{8}$$

Definition: Conditional Probability and Event Independence

Conditional Probability: The conditional probability of an event A given that event B has already occurred ($P(B) > 0$) is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \tag{9}$$

Independence of Events: Two events A and B are statistically **independent** if and only if:

$$P(A \cap B) = P(A) \cdot P(B) \iff P(A \mid B) = P(A) \quad (\text{provided } P(B) > 0) \tag{10}$$

Mutual Independence of n Events: Events $A_1, A_2, \ldots, A_n$ are mutually independent if for every sub-collection of k events ($2 \leq k \leq n$):

$$P\left(\bigcap_{j=1}^{k} A_{i_j}\right) = \prod_{j=1}^{k} P(A_{i_j}) \tag{11}$$

Crucial Exam Distinction: Pairwise independence ($P(A_i \cap A_j) = P(A_i)P(A_j)$ for all $i \neq j$) does **not** imply mutual independence!

Law of Total Probability and Bayes' Theorem

Let $\{B_1, B_2, \ldots, B_n\}$ form a **partition** of the sample space Ω such that:

$$\bigcup_{i=1}^{n} B_i = \Omega, \quad B_i \cap B_j = \emptyset \ (\forall i \neq j), \quad \text{and } P(B_i) > 0 \ \forall i \tag{12}$$

Law of Total Probability: For any arbitrary test event $A \subseteq \Omega$:

$$P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(B_i) \cdot P(A \mid B_i) \tag{13}$$

Bayes' Theorem (Posterior Probability): For any specific hypothesis B_k given evidence

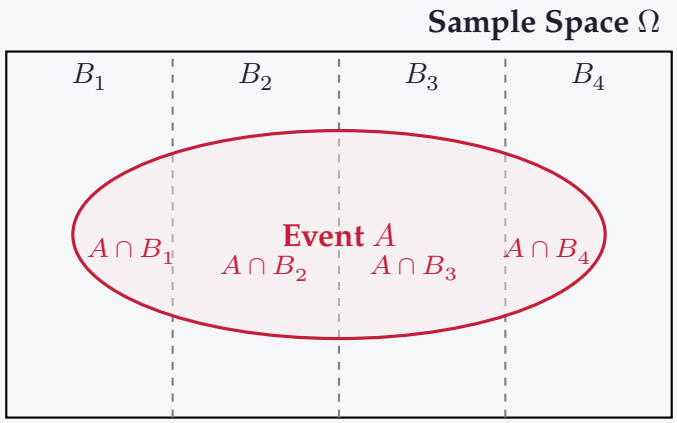
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A:

$$P(B_k \mid A) = \frac{P(B_k \cap A)}{P(A)} = \frac{P(B_k) \cdot P(A \mid B_k)}{\sum_{i=1}^{n} P(B_i) \cdot P(A \mid B_i)} \tag{14}$$

where $P(B_k)$ is the *prior probability*, $P(A \mid B_k)$ is the *likelihood*, and $P(B_k \mid A)$ is the *posterior probability*.

TikZ Schematic: Sample Space Partitioning and the Law of Total Probability



$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) + P(A \cap B_4) = \sum_{i=1}^{4} P(B_i)P(A \mid B_i)$$

2. Discrete Random Variables, PMF and CDF

Definition: Discrete Random Variable

A **random variable (RV)** X is a real-valued measurable function mapping the sample space into the real line:

$$X : \Omega \longrightarrow \mathbb{R} \tag{15}$$

A random variable is called **discrete** if its range $R_X = X(\Omega)$ is a finite or countably infinite set $\{x_1, x_2, x_3, \dots\}$.

2.1 Probability Mass Function (PMF)

The **Probability Mass Function (PMF)** assigns probabilities directly to individual realization values x_i:

$$p_X(x_i) = P(X = x_i) = P(\{\omega \in \Omega : X(\omega) = x_i\}) \tag{16}$$

Every valid discrete PMF must satisfy two fundamental conditions:

1. **Non-negativity:** $p_X(x_i) \geq 0 \quad \forall x_i \in R_X$
2. **Unit Normalization:** $\sum_i p_X(x_i) = 1$

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2.2 Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function (CDF)**, denoted by $F_X(x)$, defines the probability that the random variable X takes on a value less than or equal to a specified threshold $x \in \mathbb{R}$:

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} p_X(x_i) \tag{17}$$

Mathematical Properties of the Discrete CDF

The CDF $F_X(x)$ of any discrete random variable satisfies:

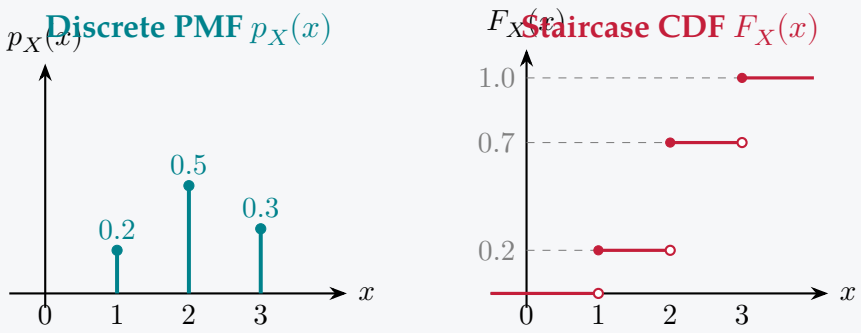
1. **Boundary Limits:** $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to +\infty} F_X(x) = 1$
2. **Monotonic Non-Decreasing:** If $a < b$, then $F_X(a) \leq F_X(b)$.
3. **Right-Continuity:** $\lim_{h \to 0^+} F_X(x+h) = F_X(x)$
4. **Step Discontinuities (Jump Heights):** At every point x_k with non-zero mass $p_X(x_k) > 0$, $F_X(x)$ exhibits a jump discontinuity of magnitude equal to the PMF:

$$P(X = x_k) = F_X(x_k) - F_X(x_k^-) = F_X(x_k) - \lim_{\epsilon \to 0^+} F_X(x_k - \epsilon) \tag{18}$$

5. **Interval Probabilities:**

$$P(a < X \leq b) = F_X(b) - F_X(a) \tag{19}$$
$$P(a \leq X \leq b) = F_X(b) - F_X(a^-) = F_X(b) - F_X(a) + P(X = a) \tag{20}$$

TikZ Illustration: Discrete PMF Stems and Corresponding Staircase CDF



3. Expectation, Moments, Variance, Covariance and Correlation

Definition: Mathematical Expectation of a Discrete RV

Let X be a discrete random variable with PMF $p_X(x)$. The **expected value** (or mathematical expectation / first raw moment) of X is defined as:

$$E[X] = \mu = \sum_{x \in R_X} x \cdot p_X(x) \tag{21}$$

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provided that the series converges absolutely: $\sum_x |x| p_X(x) < \infty$. If the series diverges, the expectation does not exist.

3.1 Properties of Mathematical Expectation

For arbitrary constants $a, b \in \mathbb{R}$ and measurable functions $g(X), h(X)$:

1. **Expectation of a Constant:** $E[c] = c \sum p_X(x) = c$.
2. **Linearity of Expectation:** For any random variables X and Y (independent or dependent!):
$$E[aX + bY + c] = aE[X] + bE[Y] + c \tag{22}$$
3. **Law of the Unconscious Statistician (LOTUS):** For any function $g(X)$:
$$E[g(X)] = \sum_{x \in R_X} g(x) \cdot p_X(x) \tag{23}$$
4. **Product Rule for Independent RVs:** If X and Y are independent:
$$E[X \cdot Y] = E[X] \cdot E[Y] \tag{24}$$

3.2 Moments: Raw Moments and Central Moments

- **r-th Raw Moment (μ'_r):** Expectation about the origin:
$$\mu'_r = E[X^r] = \sum_x x^r p_X(x), \quad \mu'_1 = \mu = E[X] \tag{25}$$
- **r-th Central Moment (μ_r):** Expectation about the mean μ:
$$\mu_r = E[(X - \mu)^r] = \sum_x (x - \mu)^r p_X(x) \tag{26}$$

  Specifically:

$$\mu_0 = E[(X - \mu)^0] = 1 \tag{27}$$
$$\mu_1 = E[X - \mu] = E[X] - \mu = 0 \quad \text{(Always zero!)} \tag{28}$$
$$\mu_2 = E[(X - \mu)^2] = \text{Var}(X) = \sigma^2 \tag{29}$$
$$\mu_3 = E[(X - \mu)^3] = \mu'_3 - 3\mu'_2\mu + 2\mu^3 \quad \text{(Skewness measure)} \tag{30}$$
$$\mu_4 = E[(X - \mu)^4] = \mu'_4 - 4\mu'_3\mu + 6\mu'_2\mu^2 - 3\mu^4 \quad \text{(Kurtosis measure)} \tag{31}$$

Variance and Standard Deviation

The **variance** of X, denoted $\text{Var}(X)$ or σ^2, measures the degree of dispersion around the mean:

$$\text{Var}(X) = \sigma^2 = E\left[(X - E[X])^2\right] = E[X^2] - (E[X])^2 = \mu'_2 - (\mu'_1)^2 \tag{32}$$

Fundamental Properties of Variance:

1. $\text{Var}(X) \geq 0$, and $\text{Var}(X) = 0 \iff X = c$ (constant with probability 1).

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2. $\text{Var}(aX+b) = a^2\,\text{Var}(X)$. Adding a constant shifts position without altering spread!
3. Standard deviation is $\sigma = \sqrt{\text{Var}(X)}$, which retains the original units of X.

Definition: Covariance and Correlation Coefficient

Covariance: For two random variables X and Y with means μ_X and μ_Y, their **covariance** is:

$$\text{Cov}(X,Y) = \sigma_{XY} = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - \mu_X\mu_Y \tag{33}$$

Pearson Correlation Coefficient (ρ): Normalized measure of linear dependence:

$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y} = \frac{E[XY]-E[X]E[Y]}{\sqrt{\text{Var}(X)\,\text{Var}(Y)}} \tag{34}$$

3.3 Proof that $-1 \le \rho \le 1$

Rigorous Derivation: Cauchy-Schwarz Bound on Correlation

Consider the standardized variables $U = \frac{X-\mu_X}{\sigma_X}$ and $V = \frac{Y-\mu_Y}{\sigma_Y}$. Clearly $E[U] = E[V] = 0$, $E[U^2] = E[V^2] = 1$, and $E[UV] = \rho$. For any real scalar $t \in \mathbb{R}$, define the quadratic form:

$$Q(t) = E\left[(tU+V)^2\right] \ge 0 \quad \text{(since the square of a real quantity is non-negative)} \tag{35}$$

Expanding the quadratic expectation:

$$\begin{aligned} Q(t) &= E\left[t^2U^2 + 2tUV + V^2\right] \\ &= t^2E[U^2] + 2tE[UV] + E[V^2] \\ &= t^2(1) + 2t\rho + 1 \ge 0 \quad \forall t \in \mathbb{R} \end{aligned} \tag{36}$$

For a quadratic $At^2 + Bt + C \ge 0$ with $A = 1 > 0$ to remain non-negative for all real t, its discriminant $\Delta = B^2 - 4AC$ must be non-positive:

$$\Delta = (2\rho)^2 - 4(1)(1) \le 0 \implies 4\rho^2 - 4 \le 0 \implies \rho^2 \le 1 \tag{37}$$

Taking square roots yields:

$$-1 \le \rho \le 1 \tag{38}$$

- $\rho = +1$: Perfect positive linear relationship ($Y = aX + b$ with $a > 0$).
- $\rho = -1$: Perfect negative linear relationship ($Y = aX + b$ with $a < 0$).
- $\rho = 0$: X and Y are **uncorrelated**. Note: Independent $\implies$ Uncorrelated, but Uncorrelated does not imply Independent!

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3.4 Variance of a Sum of Random Variables

For n random variables $X_1, X_2, \dots, X_n$:

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i) + 2 \sum_{1 \le i < j \le n} \operatorname{Cov}(X_i, X_j) \tag{39}$$

For two variables:

$$\operatorname{Var}(aX \pm bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) \pm 2ab \operatorname{Cov}(X, Y) \tag{40}$$

If X and Y are independent (or merely uncorrelated so that $\operatorname{Cov}(X, Y) = 0$):

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X - Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \tag{41}$$

4. Bernoulli Trials, Multinomial, Poisson Approximation & Convolutions

4.1 Bernoulli Trials and Binomial Distribution

A **Bernoulli trial** is an experiment with exactly two mutually exclusive outcomes: *Success* (S) with probability p, and *Failure* (F) with probability $q = 1 - p$. In a sequence of n independent and identical Bernoulli trials:

Binomial Distribution Bin(n, p)

The random variable X, representing the total number of successes in n trials, follows the Binomial distribution:

$$P(X = k) = \binom{n}{k} p^k q^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}, \quad k \in \{0, 1, 2, \dots, n\} \tag{42}$$

Statistical parameters:

$$E[X] = np, \qquad \operatorname{Var}(X) = npq = np(1-p), \qquad \sigma_X = \sqrt{np(1-p)} \tag{43}$$

Since $q = 1 - p < 1$, the variance of a Binomial distribution is strictly less than its mean: $\operatorname{Var}(X) < E[X]$.

4.2 Multinomial Distribution

The **Multinomial distribution** generalizes the Binomial distribution to trials having $k \ge 2$ mutually exclusive outcomes $E_1, E_2, \dots, E_k$ with respective constant probabilities $p_1, p_2, \dots, p_k$ where $\sum_{j=1}^{k} p_j = 1$.

Definition: Multinomial PMF

In n independent repetitions, let X_j denote the number of times event E_j occurs ($\sum_{j=1}^{k} X_j = n$). The joint PMF is:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k) = \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k} \tag{44}$$

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where $x_j \in \{0, 1, \dots, n\}$ and $\sum_{j=1}^{k} x_j = n$. **Moments and Covariances:**

$$E[X_j] = np_j, \qquad \text{Var}(X_j) = np_j(1-p_j), \qquad \text{Cov}(X_i, X_j) = -np_ip_j \quad (i \neq j) \tag{45}$$

Key Insight: Covariance is strictly negative because an increase in occurrences of event E_i constrains the remaining trials available for event E_j.

4.3 Poisson Approximation to the Binomial Distribution (Law of Rare Events)

When the number of trials n is very large ($n \to \infty$) and the success probability p is very small ($p \to 0$) such that the average occurrence rate $\lambda = np$ remains a finite positive constant:

Rigorous Derivation: Poisson Limit Theorem

Starting from the Binomial PMF:

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1-\frac{\lambda}{n}\right)^{n-k} \tag{46}$$

Rearranging terms:

$$P(X=k) = \frac{\lambda^k}{k!} \cdot \left[\frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdots \frac{n-k+1}{n}\right] \cdot \left(1-\frac{\lambda}{n}\right)^n \cdot \left(1-\frac{\lambda}{n}\right)^{-k} \tag{47}$$

Taking the limit as $n \to \infty$ for fixed k:

$$\lim_{n\to\infty} \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) = 1 \cdot 1 \cdots 1 = 1 \tag{48}$$

$$\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n = e^{-\lambda} \tag{49}$$

$$\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^{-k} = (1-0)^{-k} = 1 \tag{50}$$

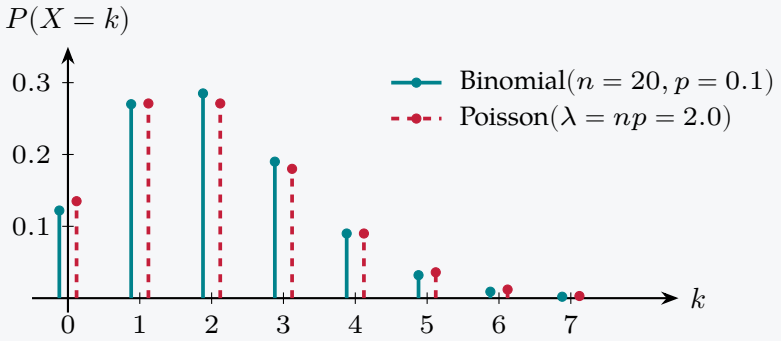
Multiplying these limits:

$$\lim_{n\to\infty} P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \dots \tag{51}$$

This is the PMF of the **Poisson distribution** with parameter $\lambda = np$. For Poisson, $E[X] = \text{Var}(X) = \lambda$.

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TikZ Plot: Binomial PMF ($n = 20, p = 0.1$) vs. Poisson PMF ($\lambda = np = 2.0$)



4.4 Infinite Sequences of Bernoulli Trials: Geometric Distribution

When independent Bernoulli trials are repeated indefinitely until the **first success** occurs:

$$P(X = k) = (1-p)^{k-1}p = q^{k-1}p, \qquad k \in \{1, 2, 3, \dots\} \tag{52}$$

Properties of the **Geometric distribution**:

$$E[X] = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2} = \frac{q}{p^2} \tag{53}$$

Memoryless Property: The geometric distribution is the *only* discrete distribution possessing memorylessness:

$$P(X > s + t \mid X > s) = P(X > t) \quad \forall s, t \in \mathbb{N} \tag{54}$$

4.5 Sums of Independent Random Variables (Discrete Convolution)

Let X and Y be independent discrete integer-valued random variables with PMFs $p_X(x)$ and $p_Y(y)$. The PMF of their sum $Z = X + Y$ is obtained via **discrete convolution**:

$$p_Z(z) = P(Z = z) = \sum_{k=-\infty}^{\infty} P(X = k, Y = z-k) = \sum_{k=-\infty}^{\infty} p_X(k) \cdot p_Y(z-k) = (p_X * p_Y)(z) \tag{55}$$

Reproductive (Convolution) Properties of Key Discrete Distributions

1. **Sum of Independent Binomials:** If $X \sim \text{Bin}(n_1, p)$ and $Y \sim \text{Bin}(n_2, p)$ share the same success parameter p:

$$Z = X + Y \sim \text{Bin}(n_1 + n_2, p) \tag{56}$$

2. **Sum of Independent Poissons:** If $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$:

$$Z = X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2) \tag{57}$$

Proof Sketch: Using the convolution formula:

$$P(Z = z) = \sum_{k=0}^{z} \frac{e^{-\lambda_1}\lambda_1^k}{k!} \cdot \frac{e^{-\lambda_2}\lambda_2^{z-k}}{(z-k)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{z!} \sum_{k=0}^{z} \binom{z}{k} \lambda_1^k \lambda_2^{z-k} = \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^z}{z!} \tag{58}$$

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5. Chebyshev's Inequality and Concentration Bounds

When the exact probability distribution of a random variable is unknown, Chebyshev's Inequality provides a guaranteed mathematical upper bound on the probability of significant deviations from the mean, requiring knowledge of only the mean μ and finite variance σ^2.

Markov's Inequality (Foundation for Chebyshev)

If Y is a non-negative random variable ($P(Y \geq 0) = 1$) with finite expectation $E[Y] < \infty$, then for any positive real constant $a > 0$:

$$P(Y \geq a) \leq \frac{E[Y]}{a} \tag{59}$$

Proof:

$$E[Y] = \sum_y y p_Y(y) = \sum_{y<a} y p_Y(y) + \sum_{y \geq a} y p_Y(y) \geq \sum_{y \geq a} y p_Y(y) \geq a \sum_{y \geq a} p_Y(y) = a P(Y \geq a) \tag{60}$$

Dividing both sides by $a > 0$ completes the proof.

Rigorous Derivation of Chebyshev's Inequality

Let X be any random variable with mean $\mu = E[X]$ and variance $\sigma^2 = \text{Var}(X) < \infty$. Define the non-negative random variable $Y = (X - \mu)^2 \geq 0$. Applying Markov's inequality to Y with threshold $a = \epsilon^2$ (where $\epsilon > 0$):

$$P\left(Y \geq \epsilon^2\right) \leq \frac{E[Y]}{\epsilon^2} \implies P\left((X-\mu)^2 \geq \epsilon^2\right) \leq \frac{E\left[(X-\mu)^2\right]}{\epsilon^2} \tag{61}$$

Since $(X - \mu)^2 \geq \epsilon^2 \iff |X - \mu| \geq \epsilon$, and $E[(X - \mu)^2] = \sigma^2$:

$$P(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2} \tag{62}$$

Substituting $\epsilon = k\sigma$ (where $k > 0$ represents the deviation in units of standard deviations):

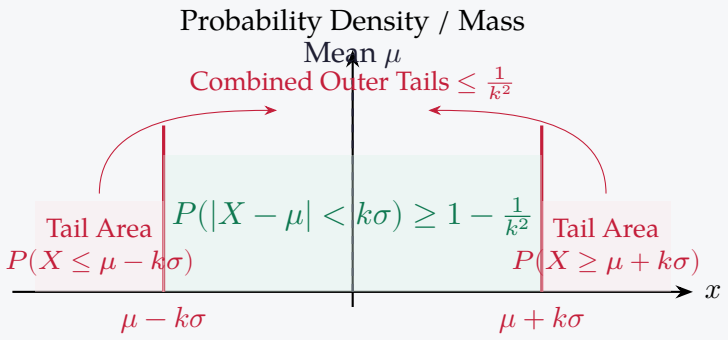
$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \tag{63}$$

Taking the complementary event yields the dual form:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \tag{64}$$

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TikZ Schematic: Geometric Interpretation of Chebyshev's Concentration Bound



5.1 Numerical Significance of Chebyshev's Bound

Spread (k)	**Interval** $[\mu - k\sigma, \mu + k\sigma]$	**Chebyshev Bound** $(P \geq 1 - 1/k^2)$	**Normal Distribution Comparison**
$k = 1$	$[\mu - \sigma, \mu + \sigma]$	$\geq 0\%$ **(Trivial bound)**	$\approx 68.27\%$
$k = 2$	$[\mu - 2\sigma, \mu + 2\sigma]$	$\geq 1 - 1/4 = 75.00\%$	$\approx 95.45\%$
$k = 3$	$[\mu - 3\sigma, \mu + 3\sigma]$	$\geq 1 - 1/9 = 88.89\%$	$\approx 99.73\%$
$k = 4$	$[\mu - 4\sigma, \mu + 4\sigma]$	$\geq 1 - 1/16 = 93.75\%$	$\approx 99.9937\%$

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Quick Revision — Module 1

Basic Probability, Discrete Random Variables and Inequalities | BS-M301

Key Formulas and Probability Identities at a Glance

Kolmogorov Axioms	$P(A) \geq 0$, $P(\Omega) = 1$, $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$ (disjoint)
Addition Law	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional Probability	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, $P(A \cap B) = P(A) \cdot P(B)$ (Independence)
Total Probability	$P(A) = \sum_{i=1}^{n} P(B_i)P(A \mid B_i)$ (Partition $\{B_i\}$)
Bayes' Theorem	$P(B_k \mid A) = \frac{P(B_k)P(A\|B_k)}{\sum_{i=1}^{n} P(B_i)P(A\|B_i)}$
Discrete Expectation	$E[X] = \mu = \sum_x x p_X(x)$, $E[aX+bY] = aE[X]+bE[Y]$
Variance Formula	$\mathrm{Var}(X) = \sigma^2 = E[X^2] - (E[X])^2$, $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$
Covariance & Correlation	$\mathrm{Cov}(X,Y) = E[XY] - \mu_X\mu_Y$, $\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y} \in [-1,1]$
Binomial Distribution	$P(X=k) = \binom{n}{k}p^k q^{n-k}$, $\mu = np$, $\sigma^2 = npq$
Poisson Approximation	$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$, $\lambda = np$, $\mu = \sigma^2 = \lambda$
Chebyshev's Inequality	$P(\|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$, $P(\|X-\mu| < k\sigma) \geq 1 - \frac{1}{k^2}$

One-Line Definitions

- **Probability Space** $(\Omega, \mathcal{F}, P)$**:** Formal triplet of sample space, σ-algebra of events, and Kolmogorov measure.
- **σ-algebra ($\mathcal{F}$):** Event collection closed under complementation and countable unions containing Ω.
- **Statistical Independence:** Condition where $P(A \cap B) = P(A)P(B)$, giving zero mutual information.
- **Random Variable:** A measurable real-valued function $X : \Omega \to \mathbb{R}$ mapping outcomes to real numbers.
- **Probability Mass Function (PMF):** Discrete assignment $p_X(x) = P(X = x)$ with non-negativity and unit sum.
- **Cumulative Distribution Function (CDF):** Non-decreasing right-continuous probability $F_X(x) = P(X \leq x)$.
- **Law of Rare Events:** Poisson convergence of Binomial distribution when $n \to \infty, p \to 0$ with $\lambda = np$ constant.
- **Memoryless Property:** Invariance condition $P(X > s+t \mid X > s) = P(X > t)$, uniquely held by Geometric distribution.
- **Uncorrelated Variables:** Variables with zero covariance ($\mathrm{Cov}(X,Y) = 0, \rho = 0$), weaker than full independence.
- **Chebyshev's Bound:** Non-parametric inequality guaranteeing at least $(1 - 1/k^2)$ probability mass within $\mu \pm k\sigma$.

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Frequently Asked / Viva Questions (Part 1)

Q1. State Kolmogorov's Axioms of Probability and derive $P(\emptyset) = 0$.

Axiom 1: $P(A) \geq 0$. Axiom 2: $P(\Omega) = 1$. Axiom 3: Countable additivity for pairwise disjoint events. Since $\Omega = \Omega \cup \emptyset$ with $\Omega \cap \emptyset = \emptyset$, Axiom 3 yields $P(\Omega) = P(\Omega) + P(\emptyset) \implies P(\emptyset) = 0$.

Q2. Distinguish between pairwise independence and mutual independence with an example.

Pairwise independence requires $P(A_i \cap A_j) = P(A_i)P(A_j)$ for all pairs. Mutual independence additionally requires the product rule across all k-tuples up to all n events. In Bernstein's example (tossing two coins, A: heads on coin 1, B: heads on coin 2, C: both same), events are pairwise independent but not mutually independent.

Q3. State and prove the Law of Total Probability.

If $\{B_1, \ldots, B_n\}$ partitions Ω, then $A = A \cap \Omega = \bigcup_{i=1}^{n} (A \cap B_i)$. Because the partitions are disjoint, the slices $(A \cap B_i)$ are mutually exclusive. By additivity, $P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(B_i)P(A \mid B_i)$.

Q4. Under what physical conditions does the Binomial distribution tend to the Poisson distribution?

When the number of independent trials $n \to \infty$, the probability of success on each trial $p \to 0$, such that the expected average number of occurrences $\lambda = np$ remains a finite positive constant (the law of rare events).

Q5. Prove that for any random variable with finite variance, $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$.

$E[aX + b] = a\mu + b$. Then $\mathrm{Var}(aX + b) = E[((aX + b) - (a\mu + b))^2] = E[(a(X - \mu))^2] = E[a^2(X - \mu)^2] = a^2 E[(X - \mu)^2] = a^2 \mathrm{Var}(X)$. Adding a constant shifts the distribution without affecting spread.

Q6. Why is covariance negative for components of a Multinomial distribution?

In a multinomial distribution, $\sum_{j=1}^{k} X_j = n$ is fixed. If event i occurs more frequently ($X_i > E[X_i]$), fewer trials remain available for other events ($X_j < E[X_j]$), creating a deterministic negative association: $\mathrm{Cov}(X_i, X_j) = -np_ip_j < 0$.

Q7. Does zero correlation ($\rho = 0$) imply independence of two random variables?

No! Independence implies uncorrelatedness ($\mathrm{Cov}(X, Y) = 0 \implies \rho = 0$), but the converse is not true in general. For example, if $X \sim U[-1, 1]$ and $Y = X^2$, then $E[X] = 0$ and $E[XY] = E[X^3] = 0$, giving $\mathrm{Cov}(X, Y) = 0$ and $\rho = 0$, yet Y is completely deterministically dependent on X!

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Frequently Asked / Viva Questions (Part 2)

Q8. State and prove Chebyshev's Inequality using Markov's Inequality.

Markov's inequality for non-negative $Y \geq 0$ states $P(Y \geq a) \leq E[Y]/a$. Defining $Y = (X-\mu)^2 \geq 0$ and setting $a = k^2\sigma^2$: $P((X-\mu)^2 \geq k^2\sigma^2) \leq E[(X-\mu)^2]/(k^2\sigma^2) = \sigma^2/(k^2\sigma^2) = 1/k^2$. Since $(X-\mu)^2 \geq k^2\sigma^2 \iff |X-\mu| \geq k\sigma$, we obtain $P(|X-\mu| \geq k\sigma) \leq 1/k^2$.

Q9. What is the reproductive property of independent Poisson random variables?

The sum of two independent Poisson variables $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ is itself a Poisson random variable with parameter $\lambda_1 + \lambda_2$. This follows from the binomial expansion within the discrete convolution sum.

Q10. Explain the memoryless property of the Geometric distribution.

$P(X > s+t \mid X > s) = P(X > t)$ for integers $s, t \geq 1$. If s consecutive failures have already occurred, the probability of requiring at least t additional trials until the first success is identical to the initial probability of requiring at least t trials. The system has no memory of past failures.

Summary Matrix: Comprehensive Comparison of Discrete Probability Distributions

Distribution	Probability Mass Function (PMF)	Mean ($E[X]$)	Variance (σ^2)	Special Characteristic
Bernoulli	$P(X=x) = p^x(1-p)^{1-x},\ x \in \{0,1\}$	p	$p(1-p)$	Single trial; basic building block.
Binomial	$P(X=k) = \binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$\sigma^2 < \mu$; independent repeated trials.
Poisson	$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!},\ k \in \mathbb{N}_0$	λ	λ	$\sigma^2 = \mu$; law of rare events.
Geometric	$P(X=k) = (1-p)^{k-1}p,\ k \in \mathbb{N}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	Only memoryless discrete distribution.
Multinomial	$P(X_1 = x_1, \dots) = \frac{n!}{\prod x_j!}\prod p_j^{x_j}$	np_j	$np_j(1-p_j)$	Negative covariance $\text{Cov} = -np_ip_j$.

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